

Cardio3D: Coronary Risk Visualisation & Prediction on a 3D Heart

Project Documentation

Track A: Cardiovascular Risk Visualisation & Prediction

1. Abstract:

Cardio3D predicts coronary artery disease and narrowing of the LAD, LCX and RCA arteries from routine clinical measurements, and displays the results on an interactive 3D heart. Trained on the 303-patient Z-Alizadeh Sani extension with nested cross-validation and Platt calibration, the models reach an out-of-fold AUC of 0.90 for CAD and 0.81 for the LAD, but only 0.72 and 0.69 for the LCX and RCA. SHAP explanations show which measurements drive each prediction. The tool is validated on one centre only and is meant for education and decision support, not diagnosis.

2. Project Overview:

Cardio3D takes a patient's routine clinical measurements and answers four questions: does this patient have coronary artery disease (CAD), and are the left anterior descending (LAD), left circumflex (LCX) or right coronary artery (RCA) narrowed by 50% or more? Each answer is a calibrated probability. The three vessel probabilities colour the matching arteries on a 3D heart built from real BodyParts3D anatomy, and a dashboard explains, measurement by measurement, why the model said what it said. Everything runs in a browser tab backed by a small Python service. The page makes no requests to any external host, so it also works offline.

3. Data & Preprocessing:

Dataset: The UCI *Extension of Z-Alizadeh Sani* file holds 303 patients and 59 columns, with no missing cells. Ages run from 30 to 86 (mean 58.9); 176 patients are recorded as male and 127 as female. Four columns are labels and the other 55 are inputs, grouped as demographics, history and risk factors, clinical examination, ECG, laboratory and echocardiography. 34 columns are numeric, and the rest are text (mostly Y/N flags). The class balance matters for reading the results:

Target	Positive Patients	Share	Meaning of Positive
CAD	216 of 303	71.3%	Catheterisation outcome is CAD (<i>Cath</i> column)
LAD	177 of 303	58.4%	LAD labelled Stenotic (at least 50% narrowing)
LCX	119 of 303	39.3%	LCX labelled Stenotic
RCA	114 of 303	37.6%	RCA labelled Stenotic

Avoiding target leakage: *LAD*, *LCX*, *RCA* and *Cath* are removed from the inputs of every model, and the overall *CAD* label is removed too. `prepare()` in `cardio/data.py` asserts that none of these names survives, and a unit test repeats the check, so a renamed column or a careless edit fails loudly instead of quietly inflating the scores.

Cleaning and Transformation: Column names are matched ignoring case and punctuation; placeholders such as "?" become missing. Numeric columns get median imputation and scaling; text columns get most-frequent imputation and one-hot encoding (unseen categories ignored). All of it sits *inside* each model pipeline, so every cross-validation fold learns its own statistics.

4. Models, Validation and Calibration:

- **Candidates.** Five SHAP-compatible models compete for each target: L2 and L1 logistic regression, random forest, extra trees and shallow gradient boosting.
- **Nested validation.** The outer loop is 3x repeated stratified 5-fold CV over all 303 patients (15 test folds per target), so every patient gets out-of-fold predictions from models that never saw them. Inside each outer training fold, the five candidates are ranked by 5-fold ROC-AUC and the winner is refit, so model choice never touches the data it is scored on.
- **Calibration.** Scores become probabilities through Platt scaling, cross-fitted for evaluation (fitted on other patients' scores).
- **Operating point.** Each target gets its own decision threshold (Youden's J on out-of-fold predictions). For evaluation, the threshold is also chosen on other patients only.
- **Uncertainty.** A patient-level bootstrap (500 resamples) gives 95% intervals for every metric.
- **Final models.** The same selection is run once on all 303 patients, and the winner is refit on everything; the Platt calibrator is fitted on all out-of-fold scores.

	CAD	LAD	LCX	RCA
Final Model	Logistic 12	Random forest	Gradient boosting	Logistic 11
Decision threshold	0.76	0.52	0.37	0.33
ROC-AOC	0.899 <i>0.86 to 0.93</i>	0.811 <i>0.76 to 0.86</i>	0.717 <i>0.66 to 0.76</i>	0.691 <i>0.64 to 0.74</i>
Accuracy	0.831 <i>0.80 to 0.86</i>	0.730 <i>0.69 to 0.77</i>	0.656 <i>0.61 to 0.70</i>	0.617 <i>0.57 to 0.66</i>
Precision	0.929 <i>0.90 to 0.96</i>	0.784 <i>0.73 to 0.84</i>	0.551 <i>0.48 to 0.62</i>	0.495 <i>0.42 to 0.57</i>
Recall	0.826 <i>0.78 to 0.87</i>	0.746 <i>0.69 to 0.79</i>	0.689 <i>0.62 to 0.75</i>	0.757 <i>0.70 to 0.82</i>
Specificity	0.843 <i>0.78 to 0.90</i>	0.709 <i>0.64 to 0.78</i>	0.634 <i>0.57 to 0.69</i>	0.533 <i>0.47 to 0.59</i>
F1	0.874 <i>0.84 to 0.90</i>	0.764 <i>0.72 to 0.80</i>	0.612 <i>0.55 to 0.67</i>	0.598 <i>0.54 to 0.65</i>
Brier Score	0.112 <i>0.09 to 0.13</i>	0.171 <i>0.15 to 0.20</i>	0.210 <i>0.19 to 0.23</i>	0.213 <i>0.19 to 0.23</i>
Calibration Error	0.053 <i>0.05 to 0.09</i>	0.058 <i>0.06 to 0.11</i>	0.081 <i>0.07 to 0.13</i>	0.054 <i>0.06 to 0.11</i>

Table 1: Out-of-fold performance on all 303 patients (mean over repeats; 95% bootstrap interval)

How to read Table 1:

- **CAD and LAD are informative.** AUC 0.90 and 0.81. At its threshold, the CAD model identifies 83% of CAD patients and correctly excludes 84% of healthy patients.

- **LCX and RCA are weak.** AUC 0.72 and 0.69, with precision near 0.5. The data support a hint, not a finding, and the interface says so on its Performance tab. We tried many model variants (L1/L2, feature selection, forests, boosting), and none lifted these two beyond about 0.74 AUC, suggesting a data limit, not a model-class limit.
- **Thresholds matter.** At a naive 0.5 cut-off, LCX and RCA recall would be only 0.41 and 0.34. The per-target thresholds trade some precision for recall; the app shows each cut-off next to the probability. CAD at 0.5 would give higher accuracy (0.85) but lower specificity (0.70).
- **Calibration.** Platt scaling lowered the calibration error for CAD, LAD and RCA (for example, CAD 0.077 to 0.053); for LCX it was slightly worse (0.068 to 0.081). With 303 patients, calibration curves stay noisy (Figure 1).
- **Nested validation removes optimism.** Selecting among the five models on all patients and reporting that score gives a higher CAD AUC (0.935) than the nested estimate (0.899). The nested figure is the one to trust.

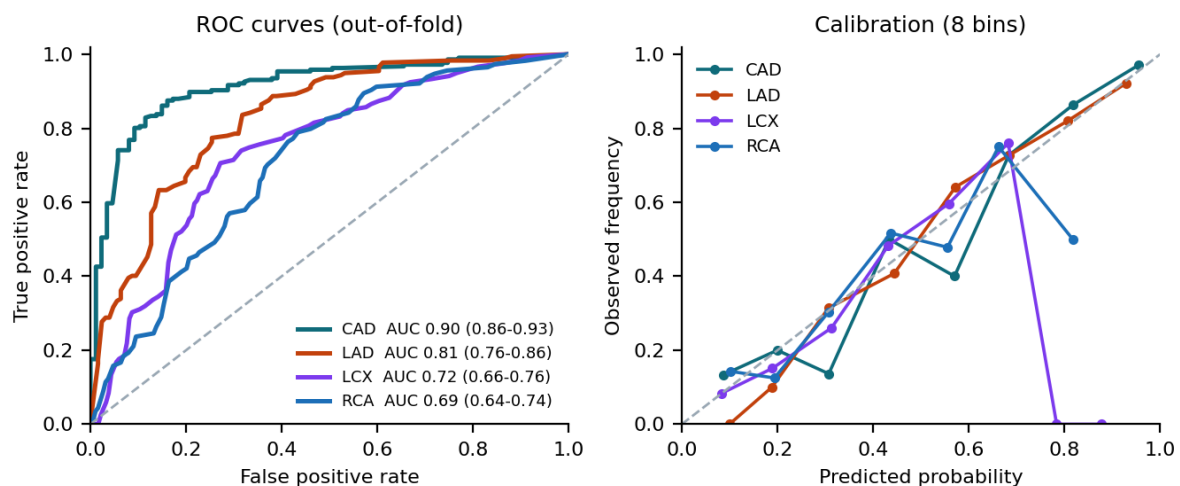


Figure 1: Out-of-fold ROC curves and calibration for the four targets (same data as Table 1)

5. Explaining a Prediction:

Each target has its own SHAP explainer (LinearExplainer for logistic models, TreeExplainer for tree models; the code lives in `cardio/explain.py`, shared by training and serving). Values of one-hot columns are summed back to the original measurement, so every input appears once. The dashboard offers three levels of explanation:

- **In words.** A sentence at the top lists the three measurements pushing the risk up and the three pulling it down, with the patient's values and units.
- **Per measurement.** Signed bars for the top 12 (or all 55), each with value, unit and share of the total absolute contribution, plus a tooltip with a plain-language description from a 55-entry glossary. Bars are in the model's own units (log-odds or probability, stated on screen); the displayed probability is the calibrated version of the same score.
- **Globally.** An Importance tab shows mean absolute SHAP over 150 patients. For overall CAD, the leading measurements are typical chest pain, age, wall-motion abnormality region, diabetes mellitus, hypertension; for RCA, they are typical chest pain, diabetes mellitus, neutrophils, age.

Selecting a heart region links the explanation to anatomy: the ventricle card shows the patient's ejection fraction and wall-motion abnormality with their share of the current explanation. Warnings appear when a value is outside the training range or when blank fields were imputed. A unit test checks that base value plus the SHAP sum reproduces the model output for all five model families.

6. The 3D Pipeline:

Anatomy: static/models/thorax.glb is built from BodyParts3D 4.0 (CC BY 4.0) by scripts/build_thorax_glb.py and slimmed by scripts/optimize_glb.py: 139 named parts, among them ventricles, atria, aorta, pulmonary trunk and veins, vena cava, ribs, sternum, spine, and the three coronary arteries as 18 (LAD), 6 (LCX) and 29 (RCA) separate segments. Decimating the skeleton took the file from 140k triangles (4 MB) to 71.5k (1.5 MB); heart and arteries are untouched.

Mapping predictions to anatomy: static/anatomy.js is the single place that says which GLB group is which target. Every segment of an artery shares one material, painted by paintVessel() on a green-to-red scale (the same HSL function as the cards and legend), so the dashboard and the 3D view cannot disagree; an automated test checks the rendered colour of each vessel against its probability for two patients. The short left main trunk is not a prediction target and stays grey.

Interaction: Drag to rotate, scroll to zoom, Front/Back/Left views, skeleton and heartbeat toggles. Hovering names the exact segment (for example "Marginal branch of right coronary artery") with its vessel's probability. Clicking an artery selects that vessel and its explanation. Clicking a heart region (ventricles, each atrium, aorta, pulmonary and venous structures) opens a card with the supplying arteries and their probabilities. Clicks and drags are told apart by a five-pixel threshold.

Performance without a dedicated GPU: The scene renders only when something changes, drops to a lower resolution while the user drags, and has a lite mode (software WebGL or ?lite=1) without antialiasing or environment maps. In headless Chromium with CPU-only software WebGL on a single core, the full scene rendered at 88 to 118 ms per frame (roughly 8 to 11 fps) across repeated runs. We have not measured physical integrated GPUs, which should be faster.

A limit worth stating: The dataset records no lesion position, so each artery shows one whole-vessel probability. A hot spot along the vessel would invent information the data does not hold.

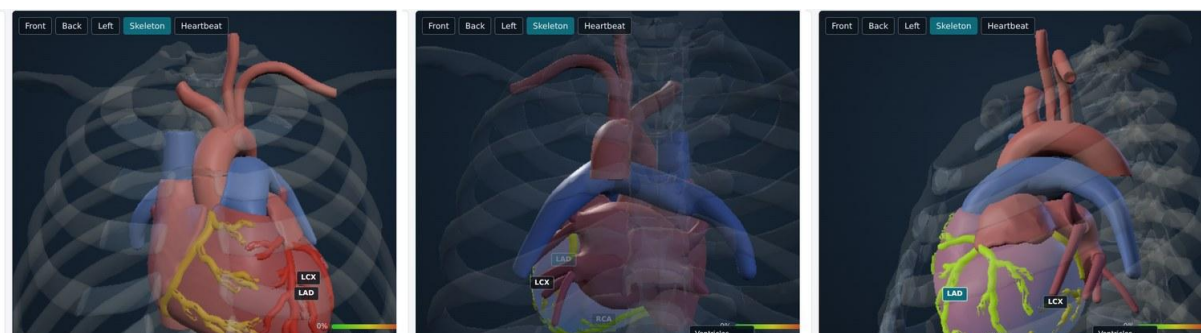


Figure 2: Front, back and left views of patient #174 (CAD-positive in the dataset). Arteries are coloured by predicted stenosis probability

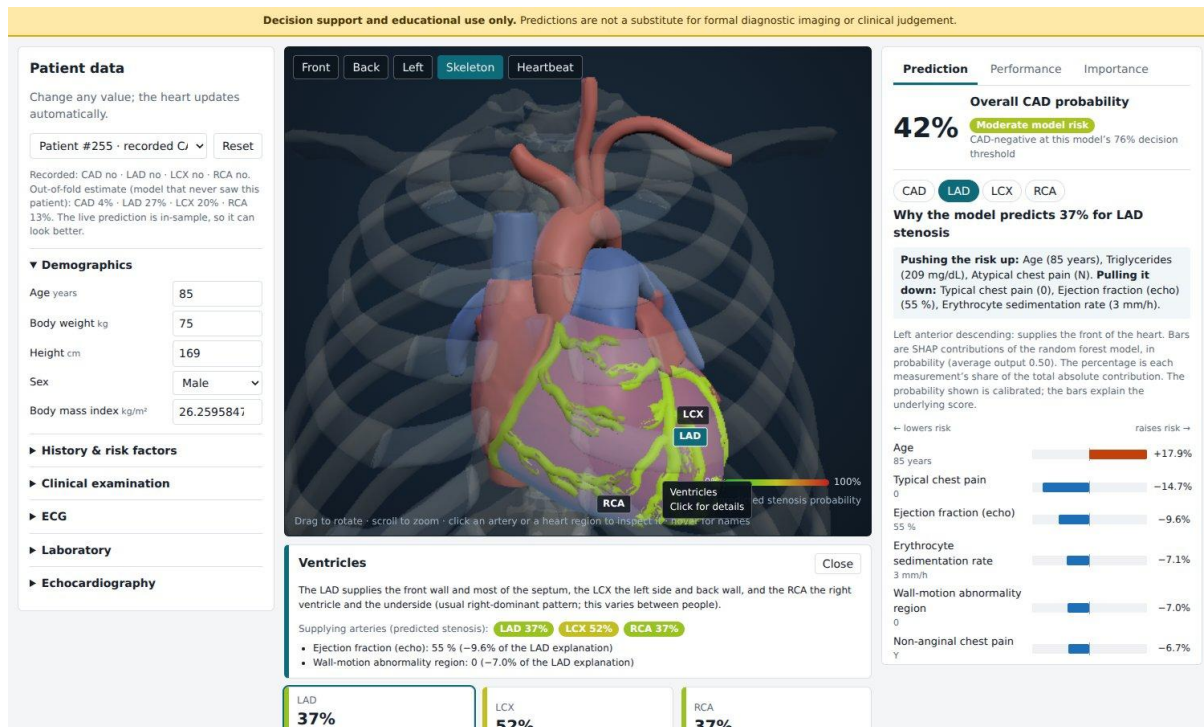


Figure 3: Dashboard with the ventricle region selected and the LAD explanation open (patient #255, with Age edited to 85 in the form)

7. System Architecture:

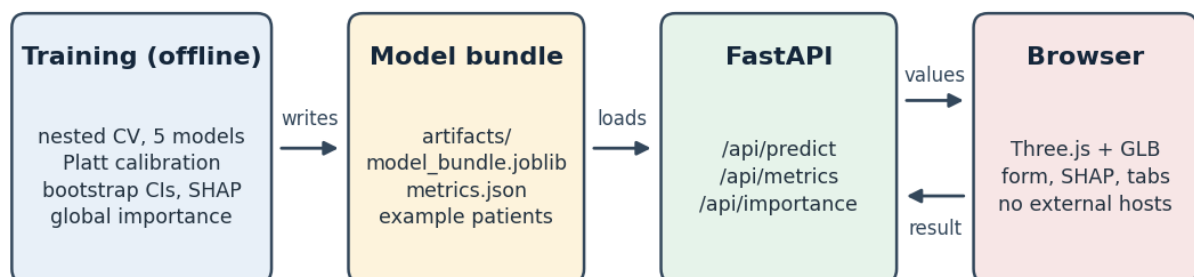


Figure 2: Training runs offline and saves one bundle; the service loads it once. Each edit in the form is debounced by 350 ms and sends one request; stale replies are discarded.

The bundle holds the five-model winners per target, Platt calibrators, metrics, per-target thresholds, global importance, the schema (with glossary), eight example patients and the SHAP background sample, plus library versions and a data hash. The service warns when the installed scikit-learn differs from the training one. Endpoints: /api/schema, /api/samples, /api/metrics, /api/importance, /api/model-info and POST/api/predict (about 0.1 s for four predictions and explanations). The form is built from the schema, so a new dataset column appears in the form, the explanation and the importance view without front-end changes. A new model is one entry in candidates(); a new structure is an entry in anatomy.js.

Tests. 12 Python tests (leakage, calibration, bounds on every metric, SHAP additivity for five model families, API behaviour, GLB-to-viewer consistency) run in about a minute on synthetic data. A 19-check browser test drives the real page in headless Chromium: GLB loading, colour-to-probability mapping, real-time repaint, clicking arteries and regions, charts, rendering cost, no console errors, no external requests.

The example patients in the app show each person's out-of-fold estimate next to the recorded outcome, and say that the live prediction for a dataset patient is in-sample.

8. Running the Project:

Requirements: Python 3.10 or newer and internet access for the first page load (Three.js is fetched from jsDelivr).

```
python -m venv .venv && source .venv/bin/activate # Windows: .venv\Scripts\Activate.ps1
pip install -r requirements.txt
uvicorn cardio.serve:app --port 8000 # then open http://localhost:8000
```

The trained models are included. To retrain (about 5 minutes on one core): `python -m cardio.train --data "data/extension of Z-Alizadeh sani dataset.xlsx"`. Tests: `python -m pytest -q tests` and, with the app running, `cd tests/e2e && npm install && npm test`. A synthetic stand-in (`python -m cardio.demo_data`) exists for software testing; the UI then shows a red SYNTHETIC banner.

9. Limitations and Next Steps:

- One centre and 303 patients, with no external validation. The 95% intervals are wide (CAD AUC 0.86 to 0.93, RCA 0.64 to 0.74).
- The four models are independent, so overall CAD can disagree with the vessel results. Probabilities are calibrated against this dataset, not against a general population.
- Risk bands (Low below 33%, Moderate below 66%, High) are display aids and are separate from each target's decision threshold, so a card can read "Moderate" and "no stenosis predicted".
- Anatomy is a generic adult thorax, not patient-specific. Browser testing used headless Chromium only. The Dockerfile has not been built.
- Any clinical use would need prospective validation, regulatory review and a clinician in the loop.

10. Sources and Credits:

Extension of Z-Alizadeh Sani Dataset, UCI Machine Learning Repository (see the dataset page for authorship and current terms). BodyParts3D 4.0, © The Database Center for Life Science, CC BY 4.0, repacked by github.com/ashemag/human-atlas. Lundberg and Lee (2017), A unified approach to interpreting model predictions, NeurIPS. scikit-learn, SHAP, FastAPI, Three.js (MIT).